Marwadi University
Faculty of Engineering and Technology
FPGA Based System Design (01CT0718)

Assignment-2

Q. No.	Question	CO	Bloom
1	Analyze the differences between behavioral and dataflow modeling styles for describing a 4-bit adder in HDL.	CO5	Analyze
2	Analyze the difference between concurrent signal assignment statements and a process statement for modeling the same combinational logic.	CO5	Analyze
3	Evaluate which modeling style (behavioral or dataflow) is more suitable for describing a large arithmetic unit, and justify your choice	CO5	Evaluate
4	Apply structural modeling to describe a 2-bit comparator using gate-level component instantiation, and explain the event-driven simulation cycle used to simulate it.	CO5	Apply
5	Apply sequential statements and a loop construct to write an HDL description of an 8-bit shift register.	CO5	Apply
6	Apply behavioral and dataflow modeling styles to completely describe an 8-bit ALU supporting addition, subtraction, and logical operations, and compare the resulting code structures.	CO5	Apply
7	Apply delay modeling and sequential circuit design concepts to model a synchronous counter in HDL, incorporating realistic propagation and clock-to-output delays.	CO3	Apply
8	Apply structural modeling to describe a 4-bit carry look-ahead adder using gate-level component instantiation, and explain in detail how the event-driven simulation cycle processes signal updates for this design.	CO5	Apply
9	Evaluate two alternative HDL descriptions (behavioral vs. dataflow) of a 16-bit multiplier in terms of synthesis quality, readability, and simulation speed.	CO5	Evaluate
10	Apply sequential statements and loop constructs to design a universal shift register in HDL, describing the complete functional behavior for all shift and load modes.	CO5	Apply
11	Apply concurrent and process-based statements to describe a 4-to-1 multiplexer in HDL, and highlight the key differences between the two approaches.	CO5	Apply
12	Evaluate the FSM coding methodology and delay modeling approach used in a given sequential circuit design, and justify improvements to meet a specified timing constraint.	CO3	Evaluate